Math 110 — Week 10

Solutions

HW10 P1.

Idea. Subtract shadows, then normalize. On $[-1,1]$ odd integrands vanish, which kills many of the inner products.

$\|1\|^2 = 2$, so $e_1 = \frac{1}{\sqrt{2}}$. $x - \langle x, e_1\rangle e_1 = x - \frac{1}{2}\int_{-1}^{1} x\,dx = x$, and $\|x\| = \sqrt{2/3}$, so $e_2 = \sqrt{\frac{3}{2}}\,x$. $x^2 - \langle x^2, e_1\rangle e_1 - \langle x^2, e_2\rangle e_2 = x^2 - \frac{1}{2}\cdot\frac{2}{3} - 0 = x^2 - \frac{1}{3}$, with $\|x^2 - \frac{1}{3}\|^2 = \int_{-1}^{1}\left(x^4 - \frac{2}{3}x^2 + \frac{1}{9}\right)dx = \frac{2}{5} - \frac{4}{9} + \frac{2}{9} = \frac{8}{45}$. So $e_3 = \sqrt{\frac{45}{8}}\left(x^2 - \frac{1}{3}\right) = \frac{\sqrt{10}}{4}(3x^2 - 1)$. Orthonormal basis: $\frac{1}{\sqrt{2}}$, $\sqrt{\frac{3}{2}}\,x$, $\frac{\sqrt{10}}{4}(3x^2-1)$.

HW10 P2.

Idea. Perpendicular to a span means perpendicular to each spanning vector: a homogeneous linear system.

$x \in W^\perp \iff x_1 - x_2 + x_3 = 0$ and $2x_2 + x_3 = 0$. Then $x_3 = -2x_2$, $x_1 = x_2 - x_3 = 3x_2$, and x_4 is free. Basis: $(3,1,-2,0)$, $(0,0,0,1)$. ($\dim W^\perp = 4 - 2$.)

HW10 P3.

Idea. Same as Problem 2: impose $\langle p, 1+x\rangle = 0$ on a general p.

For $p = a+bx+cx^2$: $\langle p, 1+x\rangle = \int_{-1}^{1}\left(a+(a+b)x+(b+c)x^2+cx^3\right)dx = 2a+\frac{2}{3}(b+c)$. So $p \in W^\perp \iff 3a+b+c=0$. Basis: $1-3x$ ($a=1, c=0$) and $x^2 - x$ ($a=0, c=1$). Check: $(1-3x)(1+x) = 1-2x-3x^2$ integrates to $2-2=0$; $(x^2-x)(1+x) = x^3 - x$ integrates to 0. ($\dim W^\perp = 3 - 1 = 2$.)

HW10 P4.

Idea. Prove the easier identity for sums, then take complements of both sides; $(Z^\perp)^\perp = Z$ undoes each $\perp$.

First, $(X+Y)^\perp = X^\perp \cap Y^\perp$ for any subspaces: $v \perp X+Y$ iff $v \perp X$ and $v \perp Y$ (every element of $X+Y$ is $x+y$, and $X, Y \subseteq X+Y$). Apply this to $X = U^\perp$, $Y = W^\perp$ and use $(Z^\perp)^\perp = Z$ (Prop. 90): $(U^\perp + W^\perp)^\perp = U \cap W$. Taking $\perp$ of both sides and using Prop. 90 again: $U^\perp + W^\perp = (U \cap W)^\perp$.

HW10 P5.

(a) **Idea.** The $U^\perp$ part of v is invisible to inner products with the e_j.

Write $v = u + w$ with $u \in U$, $w \in U^\perp$ (Prop. 89); $P_U v = u$. Since $w \perp e_j$, $\langle v, e_j\rangle = \langle u, e_j\rangle$. As $e_1, \ldots, e_m$ is an orthonormal basis of U, $u = \sum_j \langle u, e_j\rangle e_j = \sum_j \langle v, e_j\rangle e_j$.

(b) **Idea.** Split the error into a part in $U^\perp$ that every u must keep and a part in U you can remove; Pythagoras adds them in squares.

For $u \in U$, $v - u = (v - P_U v) + (P_U v - u)$ with $v - P_U v \in U^\perp$ and $P_U v - u \in U$. By Pythagoras $\|v-u\|^2 = \|v - P_U v\|^2 + \|P_U v - u\|^2 \geq \|v - P_U v\|^2$. Equality iff $\|P_U v - u\| = 0$, i.e. iff $u = P_U v$.

HW10 P6.

Idea. Riesz: find the representing vector by testing on an orthonormal basis; its i-th coordinate is $\psi(e_i)$ (no conjugates over $\mathbb{R}$).

$\psi(f) = \int_0^2 f$ is a linear functional, so Riesz gives a unique p. Orthonormal basis of $\mathcal{P}_1(\mathbb{R})$: $e_1 = \frac{1}{\sqrt{2}}$, $e_2 = \sqrt{\frac{3}{2}}x$ (HW10 P1). $p = \psi(e_1)e_1 + \psi(e_2)e_2$ (real case, no conjugates). $\psi(e_1) = \frac{2}{\sqrt{2}} = \sqrt{2}$ and $\psi(e_2) = \sqrt{\frac{3}{2}} \cdot 2$. So $p = \sqrt{2} \cdot \frac{1}{\sqrt{2}} + 2 \cdot \frac{3}{2}x = 1 + 3x$. Check: $\langle 1, 1+3x\rangle = 2 = \int_0^2 1$ and $\langle x, 1+3x\rangle = \int_{-1}^1 3x^2 = 2 = \int_0^2 x$.

HW10 P7.

Idea. ($\Leftarrow$) If some $u \in \operatorname{range} P$ and $w \in \operatorname{null} P$ were not perpendicular, moving u slightly along w would shorten it while P still sends it to u, so P would increase a length. To make this a proof, use $\|u + tw\|^2 \geq \|u\|^2$ for all t and choose t pointing against $\langle u, w\rangle$.

($\Rightarrow$) If $P = P_U$, then $v = P_U v + (v - P_U v)$ with orthogonal pieces, so $\|v\|^2 = \|P_U v\|^2 + \|v - P_U v\|^2 \geq \|P_U v\|^2$. ($\Leftarrow$) $V = \operatorname{null} P \oplus \operatorname{range} P$ (Prop. 91). Let $u \in \operatorname{range} P$, $w \in \operatorname{null} P$, and $t \in \mathbb{F}$. Then $P(u + tw) = u$ (as $Pu = u$, $Pw = 0$), so $\|u\|^2 \leq \|u + tw\|^2 = \|u\|^2 + 2\operatorname{Re}\big(\bar{t}\langle u, w\rangle\big) + |t|^2\|w\|^2$. Take $t = -s\langle u, w\rangle$ with $s > 0$ real: $0 \leq -2s|\langle u, w\rangle|^2 + s^2|\langle u, w\rangle|^2\|w\|^2$. Divide by s and let $s \to 0^+$: $|\langle u, w\rangle|^2 \leq 0$, so $\langle u, w\rangle = 0$. Hence $\operatorname{null} P \subseteq (\operatorname{range} P)^\perp$; both have dimension $\dim V - \dim \operatorname{range} P$, so $\operatorname{null} P = (\operatorname{range} P)^\perp$. So for $v = u + w$ with $u \in \operatorname{range} P$, $w \in \operatorname{null} P = (\operatorname{range} P)^\perp$: $Pv = Pu + Pw = u = P_{\operatorname{range} P} v$; hence $P = P_{\operatorname{range} P}$.

O1.

Idea. The integral is $\|x^2 - p\|^2$ for the L^2 inner product on $[0, 1]$, so the minimizer is the orthogonal projection of x^2 onto $\mathcal{P}_1(\mathbb{R})$ (best approximation).

Orthonormal basis of $\mathcal{P}_1$ (Ex. 70): $e_1 = 1$, $e_2 = \sqrt{3}(2x - 1)$. $\langle x^2, e_1\rangle = \frac{1}{3}$, $\langle x^2, e_2\rangle = \sqrt{3}\int_0^1 x^2(2x - 1)\,dx = \frac{\sqrt{3}}{6}$. So $p = \frac{1}{3} + \frac{\sqrt{3}}{6} \cdot \sqrt{3}(2x - 1) = x - \frac{1}{6}$. Minimum: $\int_0^1 \left(x^2 - x + \frac{1}{6}\right)^2 dx = \frac{1}{180}$ (this is $\|w_3\|^2$ in Ex. 70).

O2.

Idea. Projection onto a line: $\frac{\langle v, u\rangle}{\|u\|^2}u$. $U^\perp$ is one linear equation.

$P_U v = \frac{10}{4}(1, 1, 1, 1) = \left(\frac{5}{2}, \frac{5}{2}, \frac{5}{2}, \frac{5}{2}\right)$. $U^\perp = \{x_1 + x_2 + x_3 + x_4 = 0\}$, basis $(1, -1, 0, 0), (0, 1, -1, 0), (0, 0, 1, -1)$. Distance $\|v - P_U v\| = \left\|\left(-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}\right)\right\| = \sqrt{5}$.

O3.

Idea. Split $v = u + w$ along $U \oplus U^\perp$: R keeps w and flips u, a reflection.

$Rv = v - 2P_U v = (u + w) - 2u = -u + w$, so $\|Rv\|^2 = \|u\|^2 + \|w\|^2 = \|v\|^2$ (Pythagoras). $R^2 = I - 4P_U + 4P_U^2 = I$ since $P_U^2 = P_U$.